

The Impact of ESG Practices on Financial Performance in Agribusiness

Abstract

As the global focus shifts towards sustainable and ethical business practices, understanding the impact of Environmental, Social, and Governance (ESG) factors on the financial performance of agribusinesses is crucial for shaping the industry's future. This study examines the impact of combined ESG scores and their individual components on firm performance, measured through accounting-based indicators (Return on Assets (ROA) and Return on Equity (ROE)) and a market-based measure (Tobin's Q ratio). Our findings suggest that while combined ESG scores positively impact ROA and ROE, the market does not necessarily value them relatively high, as indicated by the inverse relationship with Tobin's Q ratio. When examining individual ESG components, environmental factors are associated with higher profitability measure, and governance factors are positively valued by investors. The findings have significant policy implications, emphasizing the need for policymakers to integrate ESG considerations into regulations and incentives to promote sustainability in agribusinesses.

Keywords: agribusiness, ESG performance, financial analysis, sustainability, governance

JEL: Q14, Q56, G30, M14.

1 Introduction

In recent years, the concept of Environmental, Social, and Governance (ESG) has gained significant attention across various industries, including agribusiness. Beyond effective financial management, business leaders have recognized the need to be more responsible to the environment and society and adopt measures that improve transparency in internal organization (Koundouri et al., 2022). This study employs a balance sheet and ESG score dataset of North American agribusinesses to determine the relationship between ESG factors and the financial and market performance of these firms.

Previous studies on the impact of ESG on a company's financial performance are inconclusive. Three distinct lines of research have emerged: the first indicates a positive relationship between ESG factors and profitability and market value (Alareeni and Hamdan, 2020; Gillan et al., 2010; Velte, 2017; Balatbat et al., 2012; Ademi and Klungseth, 2022), indicating that enhancing ESG factors improve firm performance measures; the second suggests an inverse relationship (Smith et al., 2007; Moore, 2001), positing that ESG commitments may detract from a firm's performance; and the third strand finds no substantial link between ESG activities and financial outcomes (Orlitzky, 2013; Patten, 1991; Sarumpaet, 2005). However, most of the existing literature provides a broad perspective on firms in general and does not focus specifically on agribusiness, which has unique characteristics that can influence the relationship between ESG and performance. Agribusinesses differ from other industries due to their direct reliance on natural resources, their exposure to climate risks, and complex supply chains, making ESG considerations particularly relevant but potentially more difficult to integrate. Regulatory pressures, consumer expectations for sustainable food production, and environmental concerns further influence the relationship between ESG and financial performance in this sector. To fully grasp the scope of the impacts of ESG, it is essential to conduct more targeted research within specialized sectors such as agribusiness, which remains underexplored in the literature.

Using the two-way fixed effects regression model, we empirically estimate the relationship between overall ESG scores, as well as individual ESG components, and firm performance. Firm

performance is evaluated using two categories of measures: accounting-based and market-based. We include both because they capture different dimensions of financial health: accounting-based measures, such as Return on Assets (ROA) and Return on Equity (ROE), reflect a firm's internal efficiency and profitability, while market-based measures, such as Tobin's Q ratio, indicate investor perceptions and market valuation. Our findings reveal that combined ESG scores have a positive impact on ROA and ROE, but an inverse relationship with Tobin's Q ratio. For example, an agribusiness firm that invests heavily in energy-efficient technology, such as solar-powered irrigation systems or low-emission machinery, may see reduced operational costs and improved efficiency, leading to higher profitability, as reflected in ROA and ROE. However, investors may be cautious about the high upfront costs and uncertain short-term returns of these investments, leading to a lower Tobin's Q ratio. Upon dissecting the ESG components, we observe that the environmental components are positively associated with ROA and ROE, and are inversely related to Tobin's Q. Furthermore, the governance score has a positive impact on the Tobin Q, suggesting that higher governance scores are associated with higher market value relative to asset replacement cost.

This study contributes to our understanding of the impact of ESG activities on firm performance in three key ways. First, it not only examines overall ESG scores, but also evaluates the effect of each individual ESG component on firms' operational and market performance. Secondly, we specifically estimate the impact of ESG activities on agribusinesses, addressing a gap in the existing literature, which generally provides a broader perspective on firms. This study is among the first to focus on the impact of ESG factors within the agribusiness sector. Finally, the findings of this research have significant policy implications, offering valuable insight to develop targeted ESG regulations for agribusinesses and guiding policymakers in the promotion of sustainability practices in this sector.

The rest of this article is structured as follows. In the next section, we provide a review of the literature on ESG scores and their association with firm performance. This section is followed by a detailed description of our data and research methodology. We then discuss our findings and conclude with a summary and policy implications.

2 Literature review

Prior to this research, numerous empirical studies have investigated the impact of Environmental, Social, and Governance (ESG) factors on the financial performance of firms, with findings that vary considerably across the literature.

The first line of literature supports a positive relationship between ESG and financial performance. For instance, Alareeni and Hamdan (2020) study the impact of ESG disclosure on the US S&P 500-listed companies from 2009 to 2018 and report that ESG disclosure improves firms' performance measures. They further demonstrate that disclosure of environmental and corporate social responsibility (CSR) is positively correlated with Tobin's Q. Similarly, Velte (2017) examines the companies listed on the German Prime Standard (DAX30, TecDAX, MDAX) and finds that ESG performance positively impacts firms' return on assets (ROA). Similar findings are reported by Gillan et al. (2010); Barnett and Salomon (2006) and Balatbat et al. (2012), all indicating a positive impact of ESG factors on firms' accounting-based and market-based measures.

Conversely, another line of research identifies a negative relationship between ESG factors and financial performance. Smith et al. (2007) examine the effect of environmental disclosure on the profitability of firms in Malaysia and find the significant inverse relationship between the environmental score and the return on assets (ROA). The negative association between environmental and social scores with ROA and ROE is also demonstrated by Alareeni and Hamdan (2020). Moore (2001) focus exclusively on corporate social performance in eight firms from the UK supermarket industry between 1994 to 1997, reporting a negative correlation between social and financial performance.

Additionally, some studies have observed a neutral relationship between ESG factors and financial performance, indicating that ESG factors are not connected to a firm's financial and market performance. Orlitzky (2013); Eccles et al. (2012) and Patten (1991) all reported no significant evidence on the relationship between ESG and firms' performance. Similarly, Sarumpaet (2005) study the relationship between environmental and financial performance among Indonesian companies and find no significant association between environmental and financial performance of a firm.

The impact of ESG factors on a firm's financial performance is complex and requires further discussion. Most current studies offer a broad perspective on firms in general, leaving a significant gap in understanding how ESG factors specifically influence agribusinesses. This is particularly important because agribusiness firms face unique challenges, such as climate risks, resource dependency, and regulatory pressures, which can alter the way ESG factors affect their financial outcomes. Without sector-specific insights, policymakers, investors, and business leaders may struggle to make informed decisions about sustainable practices and their economic implications. To fully grasp the scope of the effects of ESG, it is essential to conduct more targeted research within specialized sectors such as agribusiness, which has so far been underexplored in the literature.

3 Data and methodology

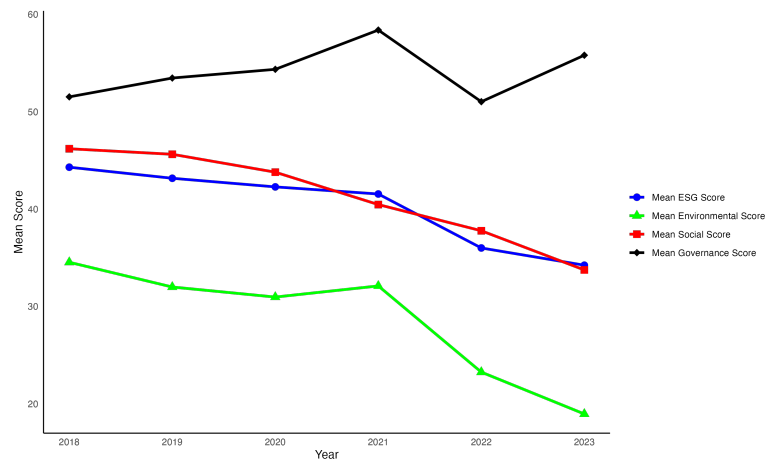
We estimate the impact of the ESG score on the financial performance of agribusiness using the LSEG Eikon datasets throughout 2018 to 2023. We filter out North American companies from the database and include only those firms that have the North American Industry Classification System (NAICS) code starting with 11. The North American Industry Classification System (NAICS) classifies companies by their primary type of business. NAICS 11 represents those firms that are primarily engaged in growing crops, raising animals, harvesting timber, and harvesting fish and other animals from a farm, ranch, or their natural habitats.

This study measures agribusiness performance based on Return on Assets(ROA), Return on Equity(ROE), and Tobin's Q ratio. ROA and ROE measure the accounting-based performance, while Tobin's Q is the market-based measure. ROE and ROA examine profitability relative to equity and assets and evaluate how efficiently equity and assets are utilized by agribusinesses. Tobin's Q is a ratio that compares the market value of a company's assets to their asset replacement cost. These three measures (ROA, ROE, and Tobin's Q) are used as dependent variables in our regression model.

The variable of interest in our study is the ESG score. We estimate the impact of the com-

bined ESG score and the individual ESG component on ROA, ROE, and Tobin's Q. The combined ESG score and individual ESG component scores for agribusinesses are obtained from the Eikon database. Figure 1 shows the trend line of the mean combined ESG score and the individual mean ESG components for agribusiness companies during the study period. Individual Environmental and Social scores are decreasing, while the Governance score is increasing over the study period. The increasing Governance score may be attributed to the positive sentiment developed in the market toward governance, which is also reflected in the regression results later in this paper. The average combined score is decreasing, as both the Environmental and Social scores are declining among the three ESG components.

Figure 1: Trend of ESG Scores for Agribusinesses



In addition to the ESG component, this study includes several control variables commonly used in this research area. The choice of these control variables is based on established practices in previous studies, which have shown their importance in analyzing the relationship between ESG factors and firm performance¹. MCAP denotes the size of the firm's market capital. We use the natural logarithm of the size of the market capital to control the size of the firm. In addition,

¹See (Alareeni and Hamdan, 2020; Velte, 2017; Gillan et al., 2010; Barnett and Salomon, 2006; Balatbat et al., 2012; Smith et al., 2007; Moore, 2001; Sarumpaet, 2005)

the beta factor is included to account for systematic risk, which measures the volatility of a firm's stock relative to the market. A negative value of Beta indicates that the firm's stock moves in the opposite direction to the market. Leverage is defined as the ratio of total debt to total assets and is used to control for unsystematic risk. The asset turnover ratio, which measures the firm's efficiency in using its assets to generate revenue, is also included as a control variable.

Table 1 reports the summary statistics of the dependent and explanatory variables used in this study. The data set has a mean ESG score of 41.08 and a standard deviation of 15.57. Individual components of ESG, Environmental, Social, and Governance scores, exhibit means of 29.79, 42.29, and 53.88, respectively. The average ROA and ROE for the agribusinesses used in this study are -0.01 and -0.07, respectively. The market-based measure, Tobin's Q ratio, shows an average value of 0.96. The mean values for beta and leverage are 0.978 and 0.17, respectively. Environmental and Governance scores have the largest standard deviations.

Table 1: Summary Statistics

Variable	Mean	SD	Median	Min	Max
ESG Score	41.08	15.57	41.95	12.68	81.11
Environmental Score	29.79	21.85	26.53	0.00	73.12
Social Score	42.29	16.87	45.30	8.17	84.07
Governance Score	53.88	22.28	53.18	9.91	92.85
ROA	-0.01	0.16	0.02	-0.88	0.45
ROE	-0.07	0.41	0.05	-1.82	0.69
Tobin's Q	0.96	1.43	0.62	0.03	11.36
Asset Turnover	0.71	0.50	0.56	0.001	1.86
log(MCAP)	20.10	1.52	20.12	15.72	22.73
Beta	0.98	0.64	0.90	-2.20	2.33
leverage	0.17	0.16	0.15	0.00	0.86

3.1 Regression model

Similar to other ESG studies (see Alareeni and Hamdan (2020); Velte (2017); Kim and Li (2021)), we employ a multivariate regression model along with time and industry-fixed effects to address endogeneity issues, which can arise when an independent variable is correlated with the error term in a regression model. Endogeneity can occur due to omitted variable bias, where relevant factors influencing both the independent and dependent variables are not included in the model; simultaneity, where the dependent variable and an independent variable mutually influence each other; or measurement error, where inaccuracies in measuring variables distort the results. To further address concerns about reverse causality, we use lagged values of ESG scores in our models to suggest that past ESG factors can influence current firm performance. This approach is particularly important as many ESG initiatives require time to demonstrate their effects on financial outcomes. By using fixed effects, we control for unobserved factors that could affect both ESG factors and firm performance, thus providing more reliable estimates of their relationship. Equation 1 and Equation 2 represent the regression models used in our analysis.

$$PERF(ROA, ROE, Tobin's Q)_{i,t} = \alpha_0 + \beta_1 ESG_{i,t-1} + \beta_2 X_{i,t} + \alpha_i + \sigma_t + \epsilon_{i,t} \quad (1)$$

$$PERF(ROA, ROE, Tobin's Q)_{i,t} = \alpha_0 + \beta_1 ENV_{i,t-1} + \beta_2 SOC_{i,t-1} + \beta_3 GOV_{i,t-1} + \beta_4 X_{i,t} + \alpha_i + \sigma_t + \epsilon_{i,t} \quad (2)$$

In equations 1 and 2, PERF, which represents the performance of the agribusiness firm, is the dependent variable in our model. We regress firm performance on the lagged ESG score to account for the possible delay in the impact of ESG factors on financial outcomes, acknowledging that ESG initiatives may require time to influence firm performance. In our study, we categorize the firm performance into two measures: accounting-based measures (ROA and ROE) and a market-based measure (Tobin's Q ratio). In equation 1, ESG denotes the overall ESG score of the firm and β_1 is an unknown parameter of interest. This parameter β_1 gives us the relationship between the overall ESG score and the firm performance. We are also interested in finding the impact of individual ESG components on the financial performance of the company. To study this impact, we use equation

2. ENV, SOC, and GOV in equation 2 represent the environmental, social, and governance score of the agribusiness firm, respectively, with β_1 , β_2 , and β_3 as the corresponding regression parameters and ϵ is the error term.

We incorporate firm size, Beta values, Asset Turnover and leverage as control variables in Equations 1 and 2, respectively. These control variables were selected based on established practices in the previous literature (Andersen and Dejoy, 2011; Alareeni and Hamdan, 2020; Fischer and Sawczyn, 2013; Han et al., 2016; Kim and Li, 2021). In the equations, X represents the set of control variables specific to the firm, α_i denotes the industry-fixed effects, σ_t represents the year-fixed effects, and $\epsilon_{i,t}$ is the error term.

4 Results and discussion

Table 2 provides the results of the multivariate regression analysis, with ROA, ROE, and Tobin's Q as dependent variable. The primary variable of interest in this study is the relationship between the overall ESG score and these dependent variables. Parameter estimates in the second and third columns of table 2 show that the ESG score has a coefficient of 0.002 with ROA and 0.001 with ROE, both of which are weakly significant. The positive coefficients for these dependent variables indicate that higher ESG scores are associated with higher ROA and ROE. This suggests that companies with better ESG practices tend to have better returns on their assets and equity. Therefore, incorporating ESG factors into agribusiness strategies might lead to more efficient use of resources and better profitability. The final column of table 2 shows that the ESG score has a coefficient of -0.020 for Tobin's Q and is statistically significant at a 99% confidence interval. This negative coefficient for Tobin's Q suggests that higher ESG scores are associated with lower Tobin's Q ratios. Keeping everything else equal, when the ESG score increases by 1 unit, the Q ratio drops by 0.02. The negative relationship with ESG score might imply that while companies with high ESG scores are efficient and profitable (as seen with ROA and ROE), the market does not necessarily value them as highly relative to their asset replacement cost. This could be due to various reasons, such as market skepticism about the financial benefits of ESG practices in the long run or potential

higher costs associated with implementing ESG practices. (Aggarwal et al., 2009; Gompers et al., 2003)

Table 2: Regression Result:

	<i>Dependent variable:</i>		
	ROA	ROE	Tobin's Q
ESG score $t-1$	0.002* (0.001)	0.001* (0.001)	-0.020*** (0.007)
Firm Size	0.021** (0.008)	0.044** (0.021)	0.383*** (0.093)
Asset Turnover	0.043*** (0.007)	0.12* (0.065)	0.20*** (0.303)
Beta	-0.002 (0.019)	-0.043** (0.043)	0.343* (0.200)
Leverage	-0.044* (0.083)	-0.353* (0.188)	-4.360*** (0.867)
Constant	-0.546*** (0.167)	-0.98** (0.378)	-5.692*** (1.747)
Observations	120	120	120
R ²	0.66	0.76	0.56
Year Fixed effect	Yes	Yes	Yes
Sub-Industry Fixed effect	Yes	Yes	Yes
<i>Note:</i> Standard errors in parenthesis *p<0.1; **p<0.05; ***p<0.01			

Table 3 presents the results of the regression analysis in which ROA, ROE, and Tobin's Q ratio are regressed with individual ESG components. There are three significant findings obtained from this result. First, environmental scores exhibit positive and significant coefficients for ROA, indicating that better environmental performance is associated with higher profitability. Agribusiness firms with better environmental scores can benefit from efficiency gains in resource use and regulatory compliance, thereby enhancing firm profitability. These results align with previous studies

by Karagozoglu and Lindell (2000); Majumdar and Marcus (2001) ; Saleh et al. (2011); Bauer et al. (2010); Bhagat and Bolton (2009) but contrast with the findings of Alareeni and Hamdan (2020). However, environmental scores are negatively associated with market valuation, as evidenced by the significant negative coefficient with Tobin's Q. Second, social scores exhibit a neutral or non-significant impact on both profitability and market-based measures, suggesting limited influence of social performance on financial and market outcome for agribusinesses in this context. Finally, governance scores show a positive association with Tobin's Q ratio, suggesting that stronger governance practices are likely valued by investors. This could be attributed to improved decision making, reduced risk, and increased investor confidence, which collectively drive higher market valuations relative to asset replacement costs.

Table 3: Regression Result:

	<i>Dependent variable:</i>		
	ROA	ROE	Tobin's Q
Environmental score $t-1$	0.001* (0.0006)	0.005 (0.0001)	-0.024*** (0.006)
Social score $t-1$	-0.001 (0.0009)	0.008 (0.0002)	-0.010 (0.009)
Governance score $t-1$	0.002 (0.0006)	0.007 (0.0001)	0.005* (0.006)
Observations	120	120	120
R2	0.67	0.70	0.60
Year Fixed effect	Yes	Yes	Yes
Sub-Industry Fixed effect	Yes	Yes	Yes
<i>Note:</i>	Standard errors in parenthesis		
	*p<0.1; **p<0.05; ***p<0.01		
<i>Controls:</i>	Firm size, Asset turnover, Beta, Leverage		

5 Conclusion

There is a growing interest about how Environmental, Social, and Governance (ESG) activities influence both the fundamental and the market performance of firms. A number of existing literatures estimate the effect of ESG activities on firms in general, yet there is a lack of empirical studies specifically targeting agribusinesses. This study employs a dataset combining balance sheet data with ESG scores for North American agribusinesses to examine the relationship between overall ESG scores, individual ESG components, and firm performance.

Using the Refinitiv Eikon database for NAICS code 11 from 2018 to 2023 and the year- and industry-fixed-effects regression model, we investigate the impact of ESG activities on firm performance. The contribution of this study is two-fold. First, we examine not only the relationship between overall ESG scores and performance, but also the impact of each ESG component on firm operational and market performance, a dimension on which the previous literature is inconclusive. Second, we specifically focus on the impact on agribusiness, addressing the research gap in the prevailing literature, which generally provides a broader perspective on firms. The findings of this study suggest that overall ESG scores have a positive impact on ROA and ROE but an inverse relationship with Tobin's Q ratio.

Regression analysis with individual ESG scores offers some interesting results. Environmental scores have a positive association with the profitability measure, suggesting that individual environmental components are associated with higher profitability. However, the social score shows a nonsignificant relationship with ROA and ROE. Furthermore, the governance score has positive coefficients for Tobin's Q, suggesting that stronger governance practices are likely valued by investors. Our findings are consistent with the findings of Karagozoglu and Lindell (2000); Bauer et al. (2010); Friedman (2007); Buallay (2019); Aggarwal et al. (2009).

This study carries significant policy implications. The impact of ESG activities on the financial and market performance of agribusinesses must be considered by policymakers to advance sustainability within this sector. The findings of this study are particularly relevant to researchers and regulators in the USA and other developed countries, as they underscore the importance of

establishing and strengthening regulations and incentives related to ESG management practices. Further, stakeholders and investors are encouraged to enhance their understanding of ESG activities and their importance, which will help them to make better investment choices.

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